



Research paper

Associations between screen time and internalizing disorder diagnoses among 9- to 10-year-olds

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ABSTRACT

Background: Children and adolescents spend an increasing amount of time with screen media. Identifying correlates of youth mental disorders has become more urgent with rates of depression, self-harm, suicide attempts, and suicide deaths rising sharply among U.S. children and adolescents after 2012. This study examined the relationship between screen time and internalizing disorders in preadolescent children between the ages of 9 and 10.

Methods: Participants were 9- and 10-year-old youth ($n = 11,780$) in the baseline of the multi-site Adolescent Brain and Cognitive Development Study (ABCD). Youth reported the number of hours a day they spent watching TV shows or movies, watching videos online, playing video games, texting, using social media, and video chatting. Youth responded to an abbreviated version of the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS-5), a semi-structured clinical interview measuring current and past symptoms of internalizing disorders using DSM-5 criteria.

Results: Youth spending 2 or more hours (vs. less than 2) a day with screen media were more likely to fit criteria for depressive disorders, self-harm, and suicidal ideation or attempts, even after adjustment for demographic covariates. For anxiety disorders, associations with digital media use (social media, texting, gaming, and online videos) were stronger than with screen time generally.

Limitations: This is a cross-sectional study utilizing retrospective screen time reports, which limits our ability to determine causality and the accuracy of the reports.

Conclusions: Preadolescents who spend more time using screens, especially digital media, are more likely to fit DSM-5 criteria for internalizing disorders.

Rates of depression, self-harm, suicide attempts, and suicide deaths have risen sharply among U.S. children and adolescents since 2012 (Keyes et al., 2019; Mercado et al., 2017; Ruch et al., 2019; Spiller et al., 2019; Twenge et al., 2019a). Given this drastic increase, along with the adverse health, legal, financial, and social outcomes associated with childhood mental health issues (Copeland et al., 2015), it is particularly important to examine how changes in environmental and societal factors might influence youth mental health.

One such notable lifestyle change is the increasing amount of time children and adolescents spend with screen media such as social media, gaming, videos, and texting. Eight- to 12-year-olds in the U.S. spend an average of 6 h of leisure time a day using screens (Common Sense Media, 2015), and 45% of adolescents say they are online “almost constantly”

(Anderson and Jiang, 2018). Time online has doubled since 2006 among U.S. adolescents (Twenge et al., 2019b). These trends illustrate the importance of examining the relationship between screen media usage and mental health issues among youth.

Quite a wealth of research has been dedicated to problematic digital media use, in both adolescents and adults. Problematic digital media usage has been found to be associated with depression, anxiety, stress, and self-esteem problems (Elhai et al., 2017; Shensa et al., 2017). Recent developments in this area include the emergence of both a theoretical construct of problematic media usage prior to adolescence and a measure of problematic media use for 4- to 11-year-olds, paving the way for expanding this research to younger children (Domoff et al., 2019, 2020). However, even relatively modest amounts of screen time may give rise

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to mental health issues without the usage being unhealthy *in itself*, and considering the ubiquity of technology in modern classrooms and households, that nuance is worth studying. Several studies have documented the association between digital screen time and mental health issues in adolescents and young adults (Duffy et al., 2019; Kelly et al., 2018; Twenge, 2019). However, the evidence is mixed, with some studies identifying no association between screen time and mental health (Coyne et al., 2020; Hollis et al., 2017). The discrepancy between these findings could be accounted for in part by methodological variations, including the operationalization and measurement of screen time, and in part by the influence of moderating variables (Twenge et al., 2020). For example, associations between screen time and mental health vary based on type of screen media activity (e.g. social media, TV, video games) and by gender (Twenge and Farley, 2020).

The current study adds to the current literature on the association between screen use and mental health issues in multiple ways. First, although several studies have examined associations between screen time and general well-being or mental health symptoms assessed in surveys (Kelly et al., 2018; Sampasa-Kanyinga and Lewis, 2015; Twenge et al., 2018), relatively little research has examined associations with mental health disorders diagnosed via clinical interviews. Perhaps as a result, this research area remains unsettled, with disagreement about the existence, size, and importance of associations between screen time and mental health, particularly around issues of measurement (Orben and Przybylski, 2019). Although child self-report of mental health symptoms has shown reasonable validity, evidence suggests that clinical interviews show superior sensitivity (Hodges, 1990), and might be methodologically advantageous to address this research question.

Second, most previous studies examining digital media use and well-being have relied on adolescent populations, with few examining the pre-adolescent years from 9 to 11, when many children receive their first smartphones (Donovan, 2016) and digital media use begins to increase (Twenge and Campbell, 2018). Given that there is some evidence suggesting a synchronous increase in the numbers of pre-adolescent children with mental health issues (Leeb et al., 2020; Lo et al., 2020), identifying whether a relationship exists between digital media use and mental health issues in this demographic is a first step to inform hypotheses regarding the role of moderating variables and potential directionality of this relationship. A better understanding of how screen time impacts prevalence of early-onset mental disorders might enable earlier and more effective mental health interventions.

In the current study, we aimed to examine associations between screen time/digital media use and DSM-5 diagnoses of internalizing disorders among pre-adolescent children. As previous studies found lower validity for parent reports (vs. youth reports) of internalizing disorders (Mesman and Koot, 2000; van de Looij-Jansen et al., 2011; Verhulst and van der Ende, 1992), we examined diagnoses based on youth reports from a semi-structured clinical interview. We drew from the first wave of the multi-site Adolescent Brain Cognitive Development Study (ABCD), which recruited 9- and 10-year-old youth.

1. Methods

1.1. Participants

The ABCD study is a multi-site, longitudinal study coordinated by the U.S. National Institutes of Health that has currently completed its baseline data collection with 9- to 10-year-old youth. The ABCD study team employed a rigorous, epidemiologically-informed school-based recruitment strategy, designed to approximate the demographics of the 21 ABCD sites and the U.S. as a whole (Volkow et al., 2018). We used ABCD Release 2.0 of the first wave of data collection; for participants with valid data on the measures below, $n = 11,780$. Average age was 9.91 (0.62); the sample was 48% female and 59% white, 9% Black, 19% Hispanic, 2% Asian, and 10% other race or ethnicity. The first wave of data collection took place in 2017 and 2018.

1.2. Measures

1.2.1. Screen time

Participants were asked 12 questions about their screen time, 6 about use “on a typical weekday” and 6 about use “on a typical weekend.” They were asked how many hours they: “watch TV shows or movies,” “watch videos (such as YouTube)?”, “Play video games on a computer, console, phone, or other device (Xbox, Play Station, iPad)?”, “Text on a cell phone, tablet, or computer (e.g., GChat, WhatsApp, etc.)?”, “visit social networking sites like Facebook, Twitter, Instagram, etc.?”, and “video chat (Skype, Facetime, etc.)?” Response choices were coded as none = 0; <30 min = 0.25; 30 min = 0.50; 1 h = 1; 2 h = 2; 3 h = 3; 4+ hours = 4. As associations with mental health disorders were similar for weekdays and weekends and the 12 questions demonstrated good internal consistency ($\alpha = 0.83$), we averaged across weekdays and weekends by multiplying the weekday time by 5, the weekend time by 2, adding the two, and then dividing by 7 to obtain a daily average.

1.2.2. Mental health disorders

Youth were assessed using a self-report computerized version of a partial Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS-5), a semi-structured interview to measure current and past symptoms of disorders using DSM-5 criteria (Barch et al., 2018; Kaufman et al., 2016). The abbreviated version of the K-SADS-5 used in ABCD included the sections for depressive disorders, generalized and social anxiety, suicidal ideation or attempts, and non-suicidal self-injurious behaviors (self-harm). Participants who met criteria on the K-SADS-5 for having a disorder in the past, who presently met criteria for a disorder, or who met criteria for currently having a disorder in partial remission were coded as a 1 for lifetime presence of the disorder, and participants who had never met criteria for a disorder were coded as a 0. This variable was utilized to obtain lifetime prevalence of these disorders. The K-SADS has historically been a reliable and valid measurement of child psychopathology (Chambers et al., 1985; Kaufman et al., 1997), and has been a gold standard research assessment for pediatric mood disorders (Alfano and Beidel, 2014; Nottelmann et al., 2001). The K-SADS-5, the most recent version updated to align with DSM-5 diagnoses, has excellent psychometric properties (Barch et al., 2018; Townsend et al., 2019).

1.2.3. Control variables

We controlled for demographic variables including child age, child race (white vs. non-white), child ethnicity (Hispanic/Latino vs. non-Hispanic/Latino), combined family income, primary caregiver education, and primary caregiver employment. To perform separate analyses by gender, we used the ABCD current gender identity variable, coding males and transgender males as male and females and transgender females as female.

1.2.4. Data analysis plan

We report data for each screen time activity separately as well as for two composites: All screen time and digital media time (including social media, gaming, texting, and videos; we excluded TV watching and video chat from digital media as TV is an older technology and video chat is similar to talking on the phone; they are thus not, strictly speaking, digital media, although they are usually considered screen time).

Given current (American Heart Association, 2018) and past (Council on Communications and Media, 2013) recommendations for screen media use, we began by comparing those spending 2 or more h/day on screen media with those spending less than 2 h/day, including all digital media use combined (including social media, gaming, texting/e-mailing, and internet use) as well as on each specific activity. We then examined the exposure-response curves for total screen time and total digital media use (comparing up to 7+ h/day of use; to balance low n 's with specificity, we combined all categories for use of less than an hour and combined those spending 3 to 5 h and 5 to 7 h). Exposure-response

curves provide information on how changes in exposure level influences disorder risk, allowing us to account for non-monotonic effects (Twenge et al., 2020). We rely primarily on relative risk (RR) and adjusted relative risk (ARR) using a univariate general linear model with covariates. Given previous evidence for gender differences in associations between screen time and well-being (Kelly et al., 2018), we performed analyses separately by gender.

2. Results

The 9- and 10-year-olds in this sample reported spending between 4 and 5 h a day using screen media, with boys spending more time than girls, especially on gaming. In contrast to samples of adolescents, in this pre-adolescent population mental health disorders were more common among boys than among girls (see Table 1).

Youth spending 2 or more hours a day with all screen media (vs. less than 2 h) were more likely to fit diagnostic criteria for depressive disorders, self-harm, and suicidal ideation or attempts, even after adjustment for demographic covariates (see Table 2). For example, girls who spent 2+ hours a day with screens (vs. less) were twice as likely to have depressive disorders. TV watching was associated with suicidal ideation or attempt among both boys and girls, with other associations for TV differing by gender. Video chat was associated with all diagnoses except anxiety disorders.

The exposure-response curves for hours of screen time and mental health disorders are shown in Supplementary Figs. 1 and 2. Girls spending 7+ hours a day on screens (vs. <1 h) were nearly four times more likely to have a depressive disorder and were twice as likely to have suicidal ideation or attempts, or engage in self-harm. Boys spending 7+ hours a day on screens (vs. <1 h) were twice as likely to have a depressive disorder and were nearly twice as likely to have suicidal ideation or attempts and non-suicidal self-injury.

The results zeroing in on newer technologies known as digital media (social media, gaming, texting, and online videos) were similar, although anxiety disorders showed significant differences with digital media use while they did not with total screen media use. The exposure-response curve for digital media was similar to that of total screen time, but with a more pronounced association for some disorders, including

Table 1

Hours per day spent on screen media and internalizing disorders measured by the K-SADS-5, boys vs. girls.

	Girls (n = 5633)	Boys (n = 6147)	F
Total screen time	4.01 (3.57) 65.3%	4.84 (3.79) 76.2%	149.10***
Total digital media	1.97 (2.15) 3.0%	2.67 (2.37) 48.7%	280.93***
TV/movies	1.25 (1.04) 22.6%	1.27 (1.05) 23.7%	1.02
Video chat	0.79 (1.02) 11.9%	0.89 (1.07) 14.3%	29.48***
Social media	0.14 (0.45) 1.7%	0.09 (0.39) 1.2%	31.08***
Gaming	0.67 (0.88) 9.9%	1.33 (1.19) 26.6%	1133.53***
Videos	0.90 (1.10) 16.4%	1.06 (1.16) 20.3%	58.81***
Texting	0.26 (0.57) 2.8%	0.19 (0.50) 2.0%	53.88***
Depressive disorders	4.8%	5.3%	1.72
Anxiety disorders	1.4%	1.3%	0.15
Suicidal ideation or attempt	7.6%	9.7%	16.42***
Self-harm	4.6%	7.4%	40.51***

Notes: For screen media rows, numbers in cells are mean (standard deviation) of screen time and % spending 2+ h/day on screen media; for mental health diagnosis rows, numbers are percent who met criteria for a depressive disorder, anxiety disorder, suicide ideation or attempt, or self-harm at some point in their lifetime.

*** = $p < .001$.

Table 2

Associations between screen time and K-SADS DSM-5 diagnoses (past and current combined), percent and *n* for use <2 h/day and 2+ h/day, unadjusted and adjusted relative risk (RR and ARR), and 95% CI.

	Total screen time	Total digital media	TV/movies	Video chat
Girls				
Depressive disorders				
Unadjusted	2.6% (1944) vs. 5.9% (3663) 2.25 (1.66, 3.03)	3.6% (3756) vs. 7.1% (1857) 1.98 (1.57, 2.50)	4.4% (4340) vs. 5.8% (1275) 1.32 (1.02, 1.71)	4.3% (4940) vs. 8.2% (674) 1.90 (1.43, 2.53)
Adjusted	2.9% (1795) vs. 5.8% (3279) 2.00 (1.48, 2.70)	3.6% (3440) vs. 7.1% (1640) 1.96 (1.53, 2.51)	4.5% (3944) vs. 5.6% (1136) 1.26 (0.95, 1.66)	4.3% (4492) vs. 8.1% (588) 1.90 (1.40, 2.56)
Anxiety disorders				
Unadjusted	1.0% (1944) vs. 1.5% (3663) 1.54 (0.91, 2.58)	0.9% (3756) vs. 2.3% (1857) 2.56 (1.64, 4.00)	1.3% (4340) vs. 1.7% (1275) 1.34 (0.82, 2.18)	1.3% (4940) vs. 1.9% (674) 1.49 (0.82, 2.69)
Adjusted	1.1% (1795) vs. 1.4% (3279) 1.26 (0.75, 2.12)	0.9% (3440) vs. 2.3% (1640) 2.57 (1.61, 4.12)	1.3% (3944) vs. 1.4% (1136) 1.09 (0.62, 1.90)	1.3% (4492) vs. 1.7% (588) 1.32 (0.68, 2.56)
Suicidal ideation or attempt				
Unadjusted	5.4% (1943) vs. 8.8% (3661) 1.63 (1.32, 2.01)	6.2% (3754) vs. 10.5% (1856) 1.69 (1.41, 2.03)	6.7% (4338) vs. 10.0% (1274) 1.49 (1.22, 1.81)	7.0% (4938) vs. 12.2% (673) 1.74 (1.39, 2.18)
Adjusted	5.5% (1794) vs. 9.1% (3279) 1.65 (1.32, 2.05)	6.4% (3439) vs. 10.9% (1640) 1.71 (1.41, 2.06)	7.2% (3943) vs. 10.1% (1136) 1.41 (1.14, 1.73)	7.2% (4491) vs. 12.7% (588) 1.77 (1.40, 2.25)
Self-harm				
Unadjusted	3.6% (1944) vs. 5.1% (3663) 1.42 (1.08, 1.86)	3.9% (3756) vs. 5.8% (1857) 1.49 (1.17, 1.91)	4.2% (4340) vs. 5.8% (1275) 1.38 (1.06, 1.80)	4.4% (4940) vs. 6.1% (674) 1.38 (1.00, 1.91)
Adjusted	3.5% (1795) vs. 5.1% (3279) 1.45 (1.09, 1.93)	3.8% (3440) vs. 6.0% (1640) 1.57 (1.22, 2.03)	4.2% (3944) vs. 5.6% (1136) 1.34 (1.01, 1.77)	4.3% (4492) vs. 6.0% (588) 1.43 (1.01, 2.03)
Boys				
Depressive disorders				
Unadjusted	3.2% (1459) vs. 5.9% (4654) 1.83 (1.35, 2.49)	3.5% (3148) vs. 7.2% (2972) 2.07 (1.65, 2.58)	4.9% (4676) vs. 6.5% (1454) 1.33 (1.06, 1.68)	4.8% (5252) vs. 8.4% (871) 1.75 (1.36, 2.24)
Adjusted	3.5% (1342) vs. 5.8% (4148) 1.66 (1.22, 2.25)	3.5% (2846) vs. 7.1% (2650) 2.02 (1.59, 2.56)	5.0% (4211) vs. 6.1% (1293) 1.22 (0.95, 1.57)	4.9% (4729) vs. 7.5% (769) 1.53 (1.16, 2.03)
Anxiety disorders				
Unadjusted	0.7% (1459) vs. 1.5% (4654) 2.19 (1.13, 4.25)	1.0% (3148) vs. 1.6% (2972) 1.64 (1.05, 2.57)	1.0% (4676) vs. 2.1% (1454) 2.12 (1.35, 3.33)	1.2% (5252) vs. 1.8% (871) 1.53 (0.89, 2.64)
Adjusted				

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Table 2 (continued)

	Total screen time	Total digital media	TV/movies	Video chat
	0.7% (1342) vs. 1.5% (4148) 2.23 (1.11, 4.47)	0.9% (2846) vs. 1.7% (2650) 1.86 (1.15, 3.00)	1.1% (4211) vs. 2.1% (1293) 1.91 (1.19, 3.06)	1.2% (4729) vs. 2.0% (769) 1.62 (0.92, 2.84)
Suicidal ideation or attempt				
Unadjusted	8.0% (1459) vs. 10.3% (4651) 1.28 (1.06, 1.56)	8.0% (3148) vs. 11.6% (2969) 1.45 (1.24, 1.69)	9.2% (4676) vs. 11.4% (1451) 1.24 (1.04, 1.47)	9.0% (5250) vs. 14.0% (870) 1.57 (1.29, 1.87)
Adjusted	8.0% (1342) vs. 10.1% (4145) 1.27 (1.03, 1.55)	7.9% (2846) vs. 11.5% (2647) 1.45 (1.23, 1.71)	9.1% (4211) vs. 11.2% (1290) 1.23 (1.02, 1.47)	8.8% (4727) vs. 14.4% (768) 1.64 (1.35, 2.00)
Self-harm				
Unadjusted	5.7% (1459) vs. 7.8% (4654) 1.37 (1.09, 1.73)	5.7% (3148) vs. 9.1% (2972) 1.60 (1.33, 1.92)	7.0% (4676) vs. 8.5% (1454) 1.22 (1.00, 1.49)	6.8% (5252) vs. 10.5% (871) 1.54 (1.24, 1.91)
Adjusted	5.8% (1342) vs. 7.9% (4148) 1.36 (1.07, 1.73)	5.7% (2846) vs. 9.2% (2650) 1.62 (1.34, 1.96)	7.0% (4211) vs. 8.6% (1293) 1.23 (0.99, 1.51)	6.8% (4729) vs. 10.7% (769) 1.57 (1.24, 1.97)

Notes: 1. *n*'s for <2 h/day vs. 2+ h/day are in parentheses. 2. RR's for which the 95% confidence interval does not include 1 are in bold. 3. Numbers in the adjusted rows include demographic controls.

depressive disorders among boys and anxiety disorders among both boys and girls (see Supplementary Figs. 3 and 4).

Among individual digital media activities, social media use was significantly associated with depressive disorders among both boys and girls and with non-suicidal self-injury and suicidal ideation or attempt among boys. Gaming was significantly associated with suicidal ideation or attempt among both boys and girls, with other associations varying by gender (see Table 3). Texting was not associated with mental health disorders among girls in analyses adjusted by covariates, though it was associated with depressive disorders and anxiety disorders among boys. Watching online videos was associated with all diagnoses except anxiety disorders among girls in adjusted analyses.

Given the small number of 9- to 10-year-olds who spent more than 2 h a day on social media (1.7% of girls and 1.2% of boys), we also compared those who spent no time on social media to those who spent some time. In analyses adjusted for covariates, social media users (vs. non-users) were more likely to have depressive disorders [7.1% (1081) vs. 4.1% (4002), ARR = 1.74 (1.34, 2.26) for girls; 6.9% (824) vs. 5.0% (4681), ARR = 1.38 (1.05, 1.83) for boys]. Associations with suicidal ideation or attempts were significant for girls [9.5% (1081) vs. 7.4% (4001), ARR = 1.29 (1.04, 1.60)] but not for boys [10.8% (822) vs. 9.4% (4680), ARR = 1.15 (0.93, 1.43)].

However, there were no significant differences among users and non-users of social media in anxiety disorders [1.4% (1081) vs. 1.3% (4002) ARR = 1.07 (0.60, 1.89) for girls; 1.6% (824) vs. 1.3% (4681) ARR = 1.21 (0.67, 2.19) for boys] or in self-harm [5.5% (1081) vs. 4.3% (4002) ARR = 1.27 (0.95, 1.69) for girls; 7.7% (824) vs. 7.3% (4681), ARR = 1.05 (0.81, 1.35) for boys].

Table 3

Associations between specific digital media activities and K-SADS DSM-5 diagnoses (past and current combined), percent and *n* for use <2 h/day and 2+ h/day, unadjusted and adjusted relative risk (RR and ARR), and 95% CI.

	Social media	Gaming	Texting	Online videos
Girls				
Depressive disorders				
Unadjusted	4.6% (5522) vs. 10.5% (95) 2.29 (1.26, 4.16)	4.6% (5059) vs. 5.9% (556) 1.29 (0.90, 1.84) 4.6% (4596) vs. 5.7% (484) 1.26 (0.86, 1.85)	4.8% (5462) vs. 3.9% (156) 0.80 (0.36, 1.77) 4.7% (4951) vs. 3.7% (132) 0.80 (0.34, 1.92)	4.2% (4691) vs. 7.5% (926) 1.77 (1.36, 2.13) 4.1% (4262) vs. 7.7% (821) 1.87 (1.42, 2.47)
Adjusted	4.6% (5004) vs. 10.7% (79) 2.20 (1.13, 4.30)	4.6% (4596) vs. 5.7% (484) 1.26 (0.86, 1.85)	4.7% (4951) vs. 3.7% (132) 0.80 (0.34, 1.92)	4.1% (4262) vs. 7.7% (821) 1.87 (1.42, 2.47)
Anxiety disorders				
Unadjusted	1.3% (5522) vs. 2.1% (95) 1.61 (0.40, 6.48)	1.2% (5059) vs. 2.7% (556) 2.24 (1.28, 3.91)	1.4% (5462) vs. 1.3% (156) 0.92 (0.23, 3.72)	1.1% (4691) vs. 2.3% (926) 2.05 (1.24, 3.78)
Adjusted	1.3% (5004) vs. 2.2% (79) 1.95 (0.49, 7.82)	1.2% (4596) vs. 2.4% (484) 2.07 (1.12, 3.84)	1.4% (4951) vs. 0.4% (132) 0.54 (0.08, 3.88)	1.2% (4262) vs. 2.1% (821) 1.73 (1.00, 2.98)
Suicidal ideation or attempt				
Unadjusted	7.6% (5519) vs. 10.5% (95) 1.39 (0.77, 2.51)	7.4% (5056) vs. 9.5% (556) 1.29 (0.98, 1.69)	7.6% (5459) vs. 7.7% (156) 1.01 (0.58, 1.76)	6.7% (4688) vs. 12.4% (926) 1.85 (1.52, 2.27)
Adjusted	7.8% (5003) vs. 10.6% (79) 1.30 (0.67, 2.52)	7.7% (4595) vs. 9.6% (484) 1.23 (0.92, 1.65)	7.9% (4950) vs. 6.8% (132) 0.86 (0.46, 1.63)	6.8% (4261) vs. 13.2% (821) 1.93 (1.57, 2.38)
Self-harm				
Unadjusted	4.5% (5522) vs. 8.4% (95) 1.88 (0.96, 3.68)	4.4% (5059) vs. 5.8% (556) 1.31 (0.91, 1.87)	4.5% (5462) vs. 6.4% (156) 1.42 (0.77, 2.62)	4.3% (4691) vs. 6.1% (926) 1.40 (1.05, 1.87)
Adjusted	4.5% (5004) vs. 7.1% (79) 1.69 (0.77, 3.68)	4.4% (4596) vs. 5.6% (484) 1.27 (0.86, 1.88)	4.5% (4951) vs. 6.5% (132) 1.51 (0.80, 2.88)	4.2% (4262) vs. 6.3% (821) 1.51 (1.12, 2.03)
Boys				
Depressive disorders				
Unadjusted	5.1% (6056) vs. 16.2% (74) 3.18 (1.87, 5.40)	4.6% (4500) vs. 7.1% (1627) 1.55 (1.24, 1.93)	5.1% (6009) vs. 13.9% (122) 2.74 (1.74, 4.31)	4.7% (4885) vs. 7.5% (1243) 1.59 (1.26, 2.00)
Adjusted	5.1% (5445) vs. 14.9% (60) 2.94 (1.59, 5.42)	4.8% (4051) vs. 6.7% (1450) 1.39 (1.11, 1.76)	5.1% (5403) vs. 11.7% (102) 2.30 (1.34, 3.97)	4.8% (4399) vs. 7.0% (1104) 1.45 (1.13, 1.87)
Anxiety disorders				
Unadjusted	1.3% (6056) vs. 1.4% (74) 1.04 (0.15, 7.35)	1.2% (4500) vs. 1.6% (1627) 1.33 (0.84, 2.12)	1.2% (6009) vs. 3.3% (122) 2.74 (1.02, 7.37)	1.1% (4885) vs. 1.8% (1243) 1.60 (0.98, 2.62)
Adjusted				

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Table 3 (continued)

	Social media	Gaming	Texting	Online videos
	1.3% (5445) vs. 1.8% (60) 1.28 (0.18, 9.05)	1.2% (4051) vs. 1.7% (1450) 1.43 (0.88, 2.30)	1.3% (5403) vs. 4.1% (102) 3.03 (1.13, 8.13)	1.1% (4399) vs. 2.0% (1104) 1.83 (1.11, 3.01)
Suicidal ideation or attempt				
Unadjusted	9.7% (6053) vs. 17.6% (74) 1.81 (1.10, 2.99)	9.0% (4500) vs. 11.7% (1624) 1.30 (1.10, 1.53)	9.6% (6006) vs. 15.6% (122) 1.62 (1.06, 2.47)	9.0% (4883) vs. 12.7% (1242) 1.42 (1.19, 1.68)
Adjusted	9.6% (5442) vs. 12.3% (60) 1.22 (0.60, 2.45)	8.9% (4051) vs. 11.7% (1447) 1.31 (1.10, 1.56)	9.5% (5400) vs. 13.4% (102) 1.44 (0.88, 2.37)	8.9% (4397) vs. 12.5% (1103) 1.41 (1.17, 1.69)
Self-harm				
Unadjusted	7.2% (6056) vs. 14.9% (74) 2.06 (1.19, 3.59)	6.6% (4500) vs. 9.3% (1627) 1.41 (1.17, 1.70)	7.3% (6009) vs. 9.8% (122) 1.35 (0.78, 2.32)	6.7% (4885) vs. 9.9% (1243) 1.48 (1.21, 1.80)
Adjusted	7.3% (5445) vs. 13.4% (60) 1.83 (0.95, 3.51)	6.7% (4051) vs. 9.3% (1450) 1.39 (1.14, 1.70)	7.3% (5403) vs. 8.8% (102) 1.21 (0.64, 2.27)	6.7% (4399) vs. 9.9% (1104) 1.47 (1.19, 1.82)

Notes: 1. n 's for <2 h/day vs. 2+ h/day are in parentheses. 2. RR's for which the 95% confidence interval does not include 1 are in bold. 3. Numbers in the adjusted rows include demographic controls.

3. Discussion

In a large, multi-site sample of 9- and 10-year-old U.S. youth from the ABCD study, more screen time was associated with meeting diagnostic criteria for past or current depressive disorders, anxiety disorders, self-harm, and suicidal ideation or attempts. For anxiety disorders, associations with digital media use (social media, texting, gaming, and online videos) were more pronounced than for total screen time including TV and video chat. Among both girls and boys, several types of digital media use were significantly associated with clinical parameters related to mood disorders and their features (e.g., depressive disorders; suicidality). In contrast to the conclusions of some previous studies that effect sizes in this area are small, we found considerable associations between screen time and internalizing disorders, including a doubling in the prevalence of several disorders from light to heavy use.

We identified these associations between screen time and mental health disorders in a pre-adolescent sample using clinical interviews, echoing previous results in adolescents and adults based on symptom questionnaires of mental health issues (Lin et al., 2016; Twenge et al., 2018). While causal claims cannot be made, our findings in this age group are noteworthy, since pre-adolescence marks a significant period of brain development (Giedd et al., 1999).

In past research, associations between digital media use and well-being have been stronger among girls, particularly for social media (Kelly et al., 2018). The pattern was more complex here, with some associations stronger for boys and some for girls. This may have occurred because the ABCD participants are pre-adolescent, and the greater prevalence of mood disorders among females does not appear until adolescence (Cohen et al., 1993; Twenge and Nolen-Hoeksema, 2002; Zahn-Waxler et al., 2008). The dynamics of social media use may also not operate in the same way among pre-adolescent youth, especially given the lower levels of use of social media in this sample and the relatively modest gender difference in social media use compared to

the larger differences found in adolescent samples (Kelly et al., 2018).

3.1. Limitations and implications

As only the first wave of data collection from the ABCD study was available, we cannot interpret the directionality of this association, nor whether third variables are operating. We were able to control for several possible covariates, but unmeasured third variables may nevertheless be involved. However, randomized controlled trials suggest a causal relationship between more frequent social media use and lower well-being (Hunt et al., 2018; Tromholt, 2016; Yuen et al., 2019), and longitudinal studies show that more frequent screen media use precedes low well-being (Babic et al., 2017; Lobel et al., 2017; Romer et al., 2017; Shakya and Christakis, 2017). Even if reverse causation is also occurring, caregivers and physicians should be aware that heavy users of digital media are more likely to experience mood disorders, self-harm, and suicidal ideation. If even part of the association is due to digital media use causing mental health disorders, caregivers and physicians might consider counseling parents about appropriate limits for the daily use of digital media. Given the associations between internalizing disorders and using social media in this population of 9- to 10-year-olds, advising a delay in the use of social media until later ages may also be prudent. Social media sites currently require users to affirm that they are 13 years of age or older, but this age limit is apparently not enforced (Castillo, 2018).

We were limited by the partial version of the K-SADS-5 given in this sample, which allowed us to examine internalizing disorders but not externalizing disorders, substance-use disorders, or schizophrenia. Future research should examine a wider array of mental health diagnoses in relation to screen time. In addition, retrospective report measures of screen time have their limitations, potentially leading to over- or under-estimates of time. That may be less of a concern in this sample in which the use of some screen activities (social media, texting) was low, limiting the potential variation in hourly estimates. Finally, we acknowledge that child self-reports of screen time may not be maximally accurate, but given that the literature reflects similar or greater validity of child reports of internalizing symptoms compared to parent reports, we expect that – similarly – children likely have equal or greater insight into their daily screen use habits than their parents (Choudhury et al., 2003; Kemper et al., 2003; Moretti et al., 1985).

3.2. Future directions

Our findings showing a relationship between digital media use and mental health among pre-adolescents provides urgency for future research to clarify explanatory factors, particularly since this relationship could be driven by multiple potential directional pathways. Based on the literature and current findings, we are unable to conclude whether media use increases risk for mental health difficulties, people with greater mental health difficulties tend towards greater media use, or moderating variables explain the relationship between mental health issues and media use. Previous research has suggested that the association between digital media use and mental health could be explained by sleep; digital media may interfere with sleep quality and duration, which can have detrimental effects on mental health (Barlett et al., 2012; Parent et al., 2016; Twenge et al., 2017). This effect could be particularly detrimental for pre-adolescents, since sleep disturbances in pre-adolescence predict future mental health and substance use issues (Gregory et al., 2005; Wong et al., 2009). Cyberbullying, a frequent occurrence on digital media, has been associated with greater feelings of loneliness and depressive symptoms (Olenik-Shemesh et al., 2012), but it is unknown whether cyberbullying is a minor or major factor in associations between digital media use and well-being. Greater time spent on screen media could detract from time spent on activities that are potentially protective factors against mental health risk, such as spending time in-person with friends and family or extracurricular

participation (Oberle et al., 2020; Twenge et al., 2018). Future research is needed to clarify the directionality of the relationship between digital media use and mental health and to identify the extent to which this relationship is driven by moderating variables. Understanding which clinical risk factors mediate the relationship between mental health and digital media use in adolescents may help illustrate what elements of digital media use are harmful or beneficial.

This is especially important to clarify given increased reliance on screen time for day-to-day activities, particularly after home confinement due to the COVID-19 crisis (Nielsen, 2020) and the increased development and implementation of digital interventions for youth mental health (Grist et al., 2017; Hollis et al., 2017). More to the point, a causal relationship, if it exists, between video chatting and mental disorder diagnoses may prove problematic for children forced to rely on online schooling during the COVID-19 pandemic. Preliminary research has revealed the existence of “Zoom fatigue,” a drained or exhausted feeling that results from spending a lot of the day using video conferencing software (Shockley et al., 2021). Zoom fatigue and other potentially problematic effects of video chatting could combine to the detriment of children in increasingly common online classes. The U.S. Census Bureau estimated in May of 2021 that about 21 million adults in the United States have children receiving “virtual/online instruction from a teacher in real time,” compared to about 28 million adults who have children receiving traditional, in-person instruction (U.S. Census Bureau, 2021). However, some research has indicated that non-recreational screen use (i.e., screen use for homework) is not associated with mental health outcomes (Babic et al., 2017). The effectiveness and potential iatrogenic effects of tele-mental healthcare also require further addressing, especially given that the COVID-19 pandemic has caused unforeseen surges in internalizing problems, suicidality, and need for care in children and adolescents around the world (Elharake et al., 2022; Samji et al., 2022). In short, future research should explore how to mitigate potentially deleterious effects of video calls on pre-adolescents as reliance on online and hybrid education and tele-mental healthcare expands in the wake of the COVID-19 pandemic.

Even outside of the influences of the pandemic, children and adolescents who have grown up in the 21st century have a vastly different cultural experience from prior generations. Later generations of youth report spending less time on adult activities such as work and dating, similar or less time on homework and extracurriculars, and more time on the Internet than earlier generations (Twenge and Park, 2019). The pervasive and normalized nature of digital media use and more general screen use may have more subtle effects on traits such as impulsivity and inattentiveness, which could especially have downstream effects on mental health in younger individuals who are developmentally more susceptible to these effects (Aboujaoude and Gega, 2021). Future research should explore potential mediating effects of specific symptoms and personality traits on the relationship between screen time and broader psychopathological constructs.

In conclusion, screen time/digital media usage is associated with internalizing disorders in youth ages 9–10, which may have implications for preventative care. Identifying transdiagnostic risk signs in youth is important to facilitate early detection. Screen time and digital media use could serve as marker of risk to identify youth at greater risk for mental health disorders, particularly internalizing disorders, and the discussions around limits for screen time should continue.

CRediT authorship contribution statement

J.M.T., T.E.J., and K.C. conceptualized the manuscript, analyzed the data, and wrote the original draft of the manuscript. L.R. revised the manuscript critically and prepared it for submission. All authors reviewed the results and approved the final version of the manuscript.

Declaration of competing interest

Dr. Joiner has received speaking honoraria from various organizations, has consulted for law firms on topics including death by suicide, social media, and psychological injury, and has authored several books. Dr. Twenge has received speaking honoraria from various organizations, has consulted for law firms about social media and mental health, and has authored several books.

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Appendix A. Supplementary data

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